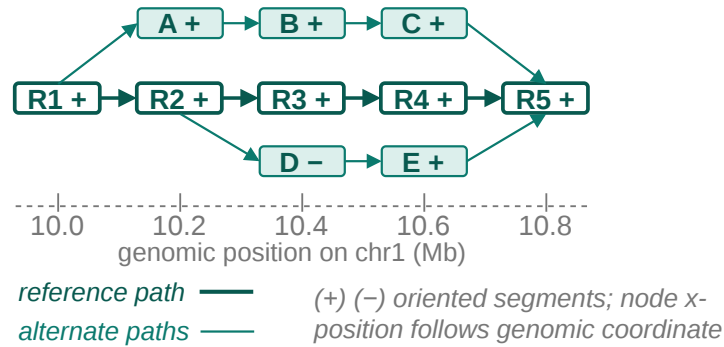


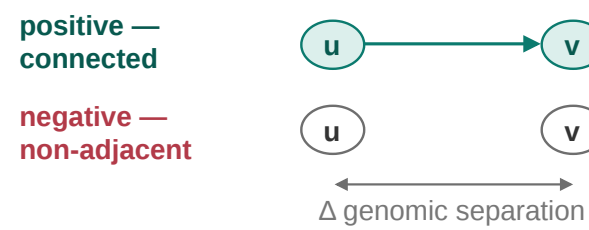
Topology-native pretraining turns an unlabeled pangenome graph into reusable segment embeddings

SELF-SUPERVISED TOPOLOGY PRETRAINING

a Population pangenome input



b Construct training pairs



Each negative is matched to a positive on Δ, so genomic distance alone does not separate the two classes.

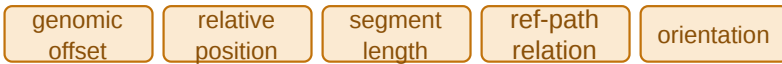
c Query-edge masking



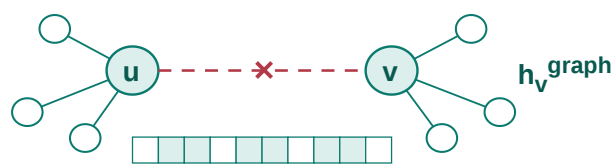
Both are deleted before message passing, so the encoder never observes the relation it is asked to score. Everything else in the 5-Mb region stays in place.

d Two information branches

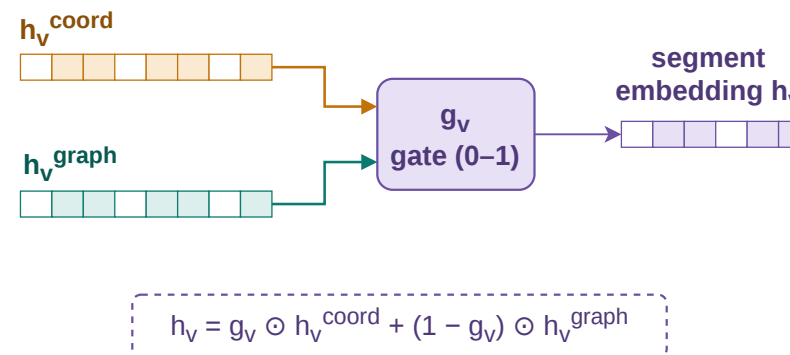
coordinate branch



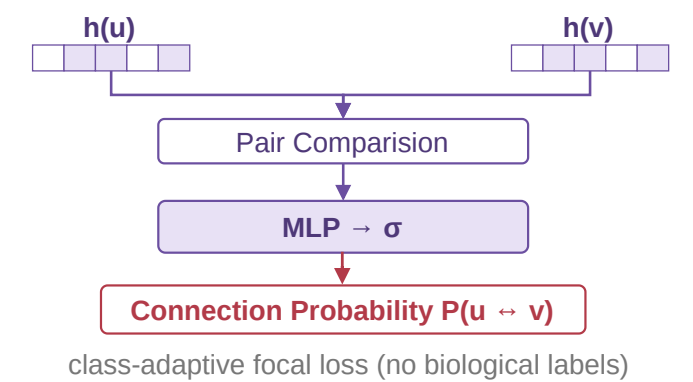
graph branch — attention over remaining edges



e Data-dependent fusion

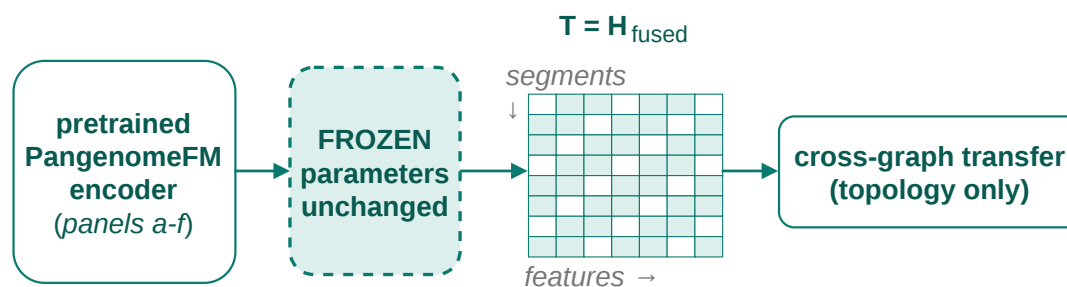


f Reconstruct the hidden edge



FREEZE AND REUSE

g Freeze the encoder, keep the embeddings



The encoder is frozen once. Transfer to another pangenome resource uses these embeddings alone, with no target-resource adaptation.

h Downstream reuse

